

SOP-SA-04 — Data SACRED Enforcement

Kode	SOP-SA-04
Pemilik	Saitama (Senior Backend Engineer) — joint @shikamaru (DBA)
Revisi	1.0 · Berlaku 2026-05-29 · Reviu per kuartal
Referensi	Tata mandate Data SACRED, ISO/IEC 27001 (audit log), tools/be-impl-checklist.md
COBIT	APO14 (Managed Data)

1. Tujuan

Menegakkan mandate **Data SACRED: never overwrite, always merge, no hard delete, tiap auto-action di-log eksplisit**. Tiap perubahan data bisa ditelusur (siapa-kapan-ngapain) — comply F-1 Backend ALMIGHTY (audit-trail lengkap, accounting-comply).

2. Ruang Lingkup

- **Berlaku:** semua mutation (create/update/delete), terutama data sensitif: money/saldo/hutang, order, payment, cart tracking.
- **Tidak berlaku:** data ephemeral murni (cache, session) yang by-design gak perlu jejak — dicatat alasannya.

3. Definisi & Istilah

- **Soft delete** — "hapus" = set kolom `deleted_at`, baris tetap ada.
- **Audit trail** — jejak append-only siapa-kapan-ngapain tiap perubahan.
- **Merge vs overwrite** — gabung perubahan (jaga riwayat) vs timpa (data lama hilang = haram).
- **Silent auto-action** — auto-settle/auto-overwrite tanpa log/notif = haram (Tata mandate).

4. Referensi

Tata mandate Data SACRED + "Auto-everything silent = haram" + PO money isolasi STRICT, ISO/IEC 27001 (logging & monitoring), [RELATIONSHIPS §4](#) (skema joint @shikamaru).

5. Tanggung Jawab (RACI)

Aktivitas	R	A	C	I
Desain audit/soft-delete di BE	Saitama	Saitama	@shikamaru (kolom audit di skema)	@kakashi
Lock skema audit (Type-1)	@shikamaru	Tata (via @kakashi)	Saitama, @kakashi	tim
Verifikasi no hard delete/overwrite	Saitama	Saitama	@kakashi (review C5)	—

6. Prosedur

6.1 Desain (joint @shikamaru)

- 6.1.1 Tiap entity mutable wajib punya kolom: `created_at`, `updated_at`, `deleted_at` (soft delete), + **kolom audit** (siapa: `created_by` / `updated_by`).
- 6.1.2 Data sensitif (money/saldo) → **append-only event log** atau versioning, bukan in-place overwrite.
- 6.1.3 **Joint sama @shikamaru**: kolom audit + constraint dikunci di skema (Type-1 → escalate via @kakashi).

6.2 Enforcement delete

- 6.2.1 **Decision point**: ada operasi delete? → **WAJIB soft delete** (`deleted_at = now()`). **Hard delete** (`DELETE FROM`) **DILARANG** (kontrol SA-C4).
- 6.2.2 Query read default filter `WHERE deleted_at IS NULL`.

6.3 Enforcement update/overwrite

- 6.3.1 **Decision point**: update tanpa nilai lama yang penting (saldo, status bayar)? → **jangan overwrite langsung** — catat perubahan di event log / versioning (merge strategy).
- 6.3.2 Concurrent edit → **optimistic locking** (kolom `version`) biar gak silent overwrite (anti lost-update).

6.4 Auto-action transparency

- 6.4.1 **Decision point**: ada auto-action (auto-settle, auto-dedup, auto-retry)? → **WAJIB log eksplisit** (siapa-trigger, kapan, kenapa) + **notif user-facing** kalau dampaknya user rasain (kontrol SA-C5). **Silent = haram**.
- 6.4.2 Money/saldo/hutang → isolasi STRICT, tiap pergerakan dana 1 entry audit.

6.5 Verifikasi

- 6.5.1 **Exit criteria**: grep migration/query — `DELETE FROM` = 0; `UPDATE` data sensitif tanpa jejak = 0; tiap auto-action ada log entry.
- 6.5.2 Test: soft-deleted row gak muncul di list, tapi masih ada di DB (riwayat terjaga).

7. Pengecualian

- **GDPR/legal hard-delete request:** kalau ada kewajiban hukum hapus permanen → **escalate Tata** 📍 (Type-1), dokumentasi alasan + persetujuan, bukan keputusan diam-diam.
- **Data ephemeral (cache/session):** boleh hard-evict, dicatat alasannya.

8. Rekaman & Retensi

Rekaman	Lokasi	Retensi
Desain audit/soft-delete (joint)	doc / log.md	Permanen
Verifikasi no hard delete (evidence grep/test)	log.md	Permanen
ADR skema audit (Type-1)	adr/NNN-*.md	Permanen
Audit trail data (di DB)	DB event log / kolom audit	Sesuai retensi data (permanen kecuali legal)

9. Indikator Kinerja (KPI)

KPI	Rumus	Target
Hard delete violation	# DELETE FROM di codebase/migration	0
Overwrite tanpa audit	# update data sensitif tanpa event log/versioning	0
Silent auto-action	# auto-action tanpa log eksplisit	0
Audit coverage entity sensitif	# entity sensitif ber-kolom-audit ÷ total	100%

10. Riwayat Revisi

Rev	Tanggal	Perubahan
1.0	2026-05-29	Terbit pertama